Phase 2

OpenStack

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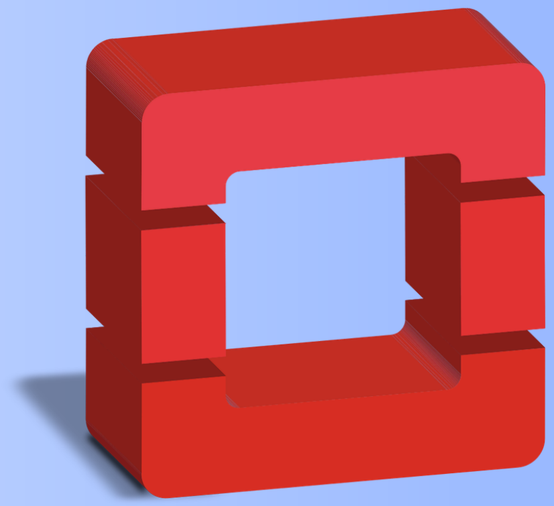
Introduction to OpenStack

What is OpenStack?

OpenStack is an open-source cloud computing platform that provides Infrastructure as a Service (IaaS).
It enables organizations to manage and automate their **compute**, **storage**, and **networking** resources through a unified platform.



openstack®
CLOUD SOFTWARE

Why OpenStack?

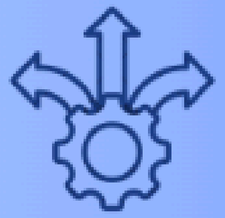
- Compatibility
- Flexible Deployment
- User-Friendly
- Cost-Effective
- Security

Key Features

Scalability

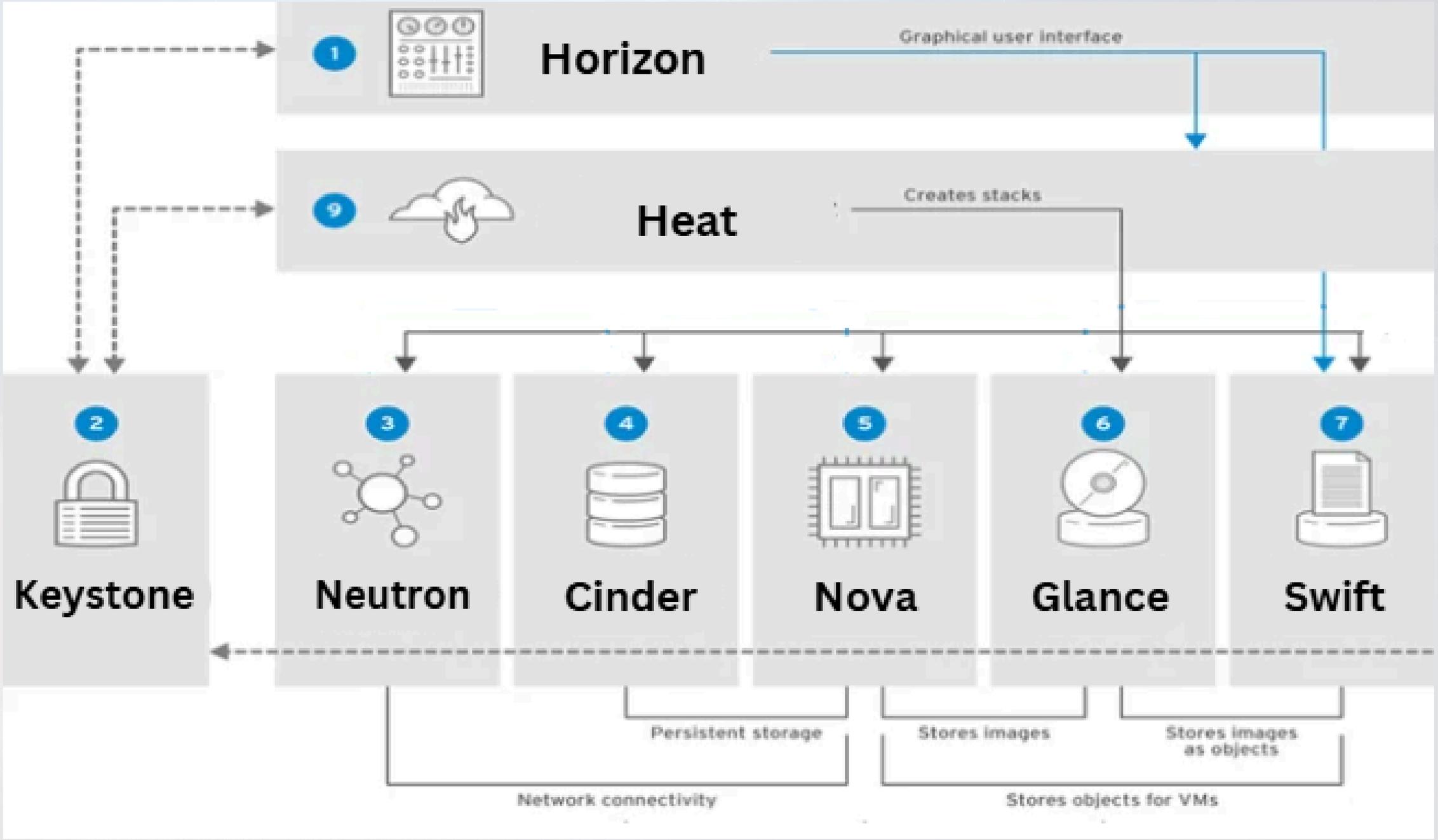
Multi-tenancy

Open-source



flexibility

OpenStack Architecture


1
Horizon
Graphical user interface
9
Heat
Creates stacks
2
Keystone
3
Neutron
4
Cinder
5
Nova
6
Glance
7
Swift
Persistent storage
Stores images
Stores images as objects
Network connectivity
Stores objects for VMs

Dashboard Service

Function:
Provides a graphical user interface (GUI) for managing OpenStack resources.

Communication:
Connects with other services through REST APIs to give users control.

Identity service

KEYSTONE

Function:
Manages user authentication and authorization.
Provides tokens to allow services to communicate.

Communication:

- Validates users and projects for accessing resources.
- Helps locate other OpenStack services.

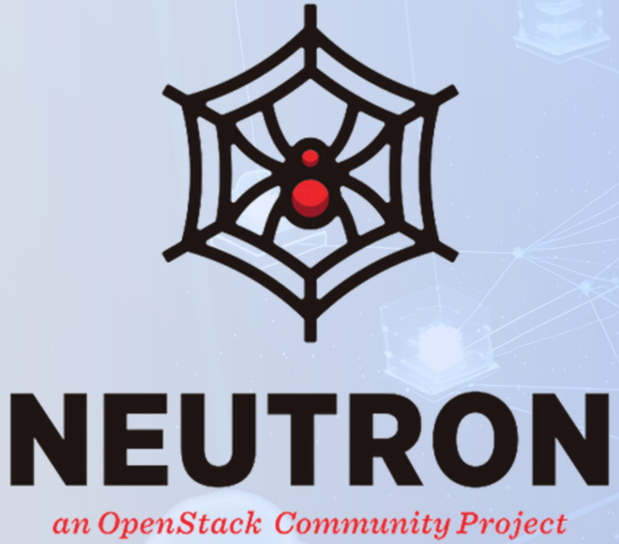
Orchestration Service

Function:
Deploys infrastructure as code (IaC) using "stacks".

Communication:

- Works with all services to provision and configure resources.
- Uses Keystone for authentication and permissions.

Networking Service



Function:
Handles virtual networking for instances, including networks, subnets, and routers.

Communication:

- Integrates with the Compute Service (Nova) to give VMs network connectivity.
- Uses plugins to manage hardware-level network configurations.

Compute Service

Function:

Manages the lifecycle of virtual machines.

Communication:

- Requests VM images from the Image Service (Glance).
- Connects VMs to networks via Neutron.
- Uses Keystone for authentication.

Image Service

Function:
Manages and stores VM images.

Communication:

- Supplies images to Nova for creating instances.
- Stores images in Object Storage (Swift) or Block Storage (Cinder).

Block Storage Service

Function:
Provides persistent storage (volumes) for VMs.

Communication:

- Attaches storage volumes to instances via Nova.
- Supports external storage systems like LVM or Ceph.

Object Storage Service

SWIFT

Function:
Provides object storage for backups, archives, or unstructured data.

Communication:

- Used by Glance to store images.
- Gives users direct access to objects through APIs.

Logical Architecture

Monitoring

Deploiment

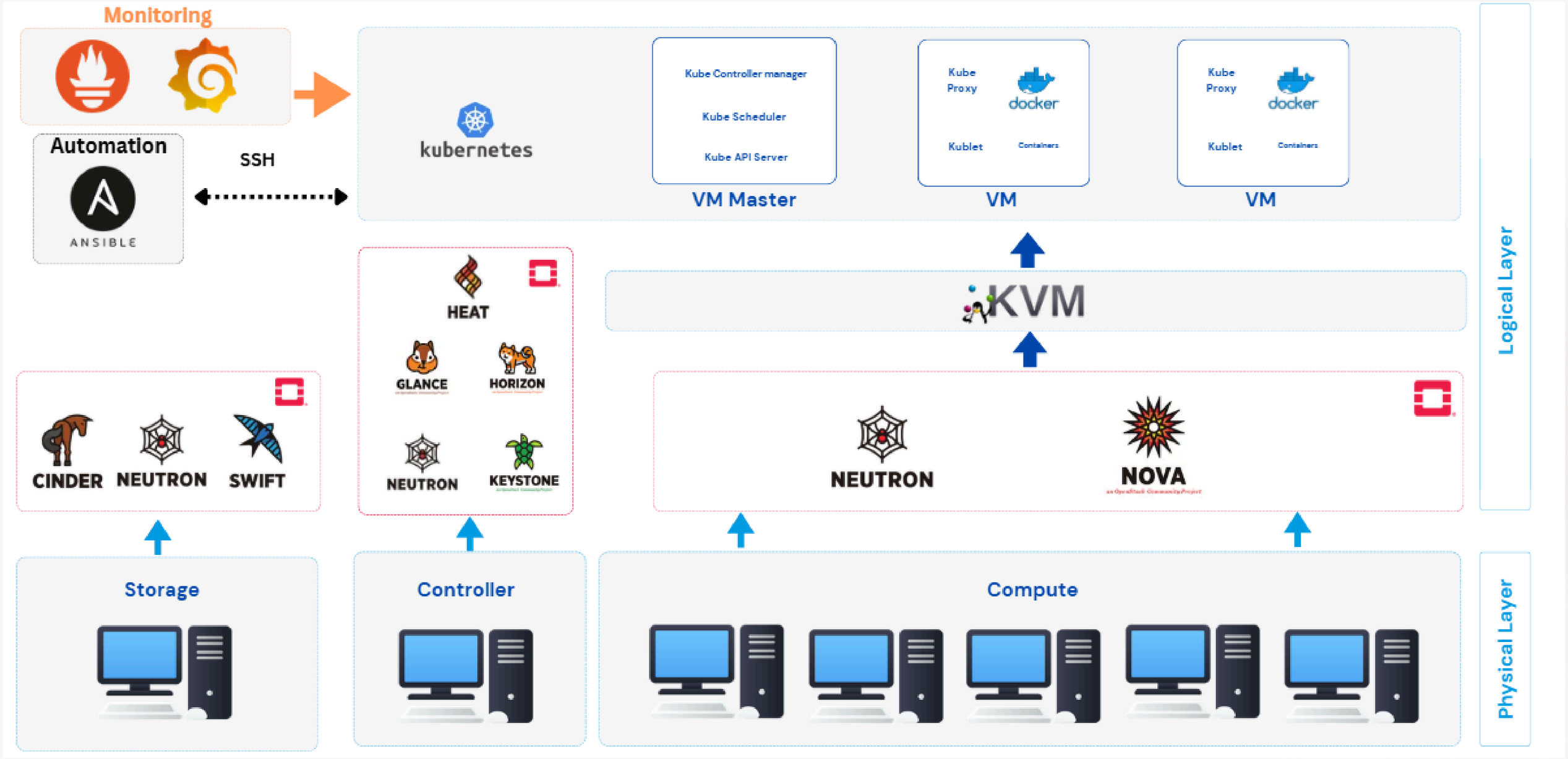
Orchestration

Infrastructure

Virtualisation

Hardware

Global Architecture


Monitoring
Automation
ANSIBLE
SSH
kubernetes
Kube Controller manager
Kube Scheduler
Kube API Server
VM Master
Kube Proxy
docker
Kublet
Containers
VM
Kube Proxy
docker
Kublet
Containers
VM
KVM
HEAT
GLANCE
HORIZON
NEUTRON
KEYSTONE
CINDER
NEUTRON
SWIFT
NEUTRON
NOVA
Logical Layer
Storage
Controller
Compute
Physical Layer

Task distribution

Islem Ghariani	Sirine Kchaou	Zeineb Korbi	Yasser Jaziri	Oussema Mrabet	Yassine Mharek	Ghada Omri
CPU : Intel i7 RAM : 32 Go Storage : 750 Go SSD	CPU: Intel ultra 7 RAM : 32 Go Storage : 252 Go SSD	CPU: Intel i7 RAM : 32Go Storage : 512 Go SDD	CPU : Intel i5 12eme RAM : 16 Go Storage : 512 Go SSD	CPU : Intel i5 11eme RAM : 32 Go Storage : 512 Go SSD	CPU : Intel Ultra 9 (24 cores) RAM : 32 Go Storage : 200 GO SSD	CPU : Intel core i5 RAM : 8 Go Storage : 250 Go SSD
Controller	**Compute**					**Storage**

OpenStack version selection

Criteria	2025.1 (Epoxy)	2024.2 (Dalmatian)	2025.2 (Flamingo)
Release Date	2025	October 2024	2025 (Expected)
End of Support	October 2026	April 2026	2027 (Expected)
Compatibility with project(2026)	Fully supported throughout the project timeline	Optimal compatibility, stable	Stable, mature, optimal for 2026
Expected Stability	Fully stable, mature	Stable and reliable for production	Expected to be highly stable
Required Services	Horizon API, Keystone, Nova, Glance, Neutron, Cinder, Heat, Swift	Horizon API, Keystone, Nova, Glance, Neutron, Cinder, Heat, Swift	Horizon API, Keystone, Nova, Glance, Neutron, Cinder, Heat, Swift
Alignement with Services Needs	Perfectly aligned with active support	Fully compatible, ideal for production	Fully compatible, stable for long-term use

chosen version

=> Flamingo (2025.2)

THANK YOU!